

Saurav Kumar

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🐙 SauravShadow

• Portfolio

• LeetCode

Summary

Software Engineer with 3 years of experience building **high-performance backend systems in C++**. Strong in **DSA, multi-threading, concurrency, and performance optimization**. Experienced in **Linux, system design, low-latency processing**, and scalable architectures. Passionate about building efficient and reliable systems.

Projects

High-Performance Video Streaming Server (C++ Systems Project)

- Designed a **modular HTTP video streaming server** in C++ with **concurrent request handling** using a custom thread pool.
- Implemented **zero-copy streaming** using `sendfile()`, reducing CPU overhead and improving throughput.
- Built **HTTP parsing, routing, and Range requests (206 Partial Content)** for efficient video seeking.
- Developed **filesystem-driven content browsing** (anime → seasons → episodes) with dynamic directory APIs.
- Containerized and deployed the service with public access via secure tunneling.
- Live:** stream.saurav-info.xyz

Multi-Threading Task Scheduler (C++ Concurrency Engine)

- Built a **high-performance task scheduler** with a **runtime-resizable thread pool**.
- Designed **REST APIs** and real-time updates via **Server-Sent Events (SSE)**.
- Simulated CPU-bound workloads to validate **thread efficiency and scheduling**.
- Live:** task.saurav-info.xyz

Experience

Keysight Technologies

Software Engineer

Bangalore, India

Oct 2023 – Present

- Developed and optimized features for a **desktop-based simulation platform** (with web integration), focusing on **high-performance C++ modules** for real-time data processing.
- Improved performance of critical components by **15%**, leveraging profiling tools and memory optimizations for low-latency execution.
- Built **multithreaded systems** to handle concurrent simulation workloads, enhancing responsiveness and throughput under heavy usage.
- Contributed to **low-level system design (LLD)** and modular architecture, improving maintainability and scalability of the simulation engine.
- Refactored legacy components into **clean, modular designs**, reducing technical debt and simplifying future feature development.
- Integrated and maintained **CI/CD pipelines** using Docker and Jenkins for streamlined build and deployment workflows.

Indian Institute of Science (IISc)

Industry Project (AON)

Bangalore, India

Jan 2023 – Jun 2023

- Processed and analyzed **large structured datasets** for risk modeling.
- Built data pipelines improving **data quality and consistency**.

Skills

Languages: C++, Python

Core: DSA, Multithreading, Concurrency, OOP, Memory Management

Systems: System Design, Low-latency Systems, **Networking (Sockets, HTTP)**

Performance: Profiling (Valgrind, Intel VTune), Debugging, Optimization

DevOps: Docker, CI/CD, GitHub Actions, Jenkins

Cloud: AWS (EC2, S3, RDS)

Databases: MySQL (Indexing, Query Optimization)

Tools: Linux, Git, **CMake**

Education

M.Tech, IISc Bangalore

2021 – 2023

B.Tech, NIT Jamshedpur

2017 – 2021